PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Interventions for improving treatment outcomes in adolescents on antiretroviral therapy with unsuppressed viral loads: a systematic review protocol
AUTHORS	Munyayi, Farai; van Wyk, Brian

VERSION 1 – REVIEW

REVIEWER	Haghighat, Roxanna University of Oxford, Social Policy and Intervention
REVIEW RETURNED	13-Mar-2021

GENERAL COMMENTS	Abstract: Some of the language should be made more precise as you write up the review, e.g. Line 15: “unsuppressed adolescents” should be made more specifically “virally unsuppressed adolescents” Methods: For a more comprehensive review, I would also look into databases of grey literature. To my knowledge, the six databases listed do not include grey literature. What tools will you use to systematically assess bias in included studies? What about tools for assessing quality of included studies? Will this SR be focusing on ALHIV care globally, or will it focus on ALHIV in a specific region? Should clarify upfront. Strengths/limitations: When you say “specifically for adolescents with unsuppressed viral loads,” does this mean your inclusion criteria will require all study participants to be unsuppressed? Intro The debate around threshold-setting for VS has also been tied to the sensitivity of available tests for detecting viral loads, which may differ across and within countries. It seems that the Casale et al and Lindsey et al papers would address the limitations of the MacPherson et al. paper. It would be good to specifically indicate where your SR would add new evidence between these two, among the other SRs listed. When looking at the existing SRs, how many of the participants in the included studies were unsuppressed at baseline? The introduction is setting me up to believe that you will only look at studies where all participants were unsuppressed at baseline, which may be *too* narrow of a focus and likely to yield low number of studies. Rationale for review: It seems that the reason for the low number of studies included in previous SRs is simply that there is a lack of sufficient studies on

	this topic. This is also the reason why many SRs have expanded eligibility criteria to include youth up to the age of 24. What does your SR do beyond these existing ones? Has there been a sufficient amount of time since the most recent SR that there is new data worth synthesizing? It would also seem important to include interventions across all levels—clinic, community, and household. But from this paragraph, it sounds like you're making the case to reduce that scope. If so, you should provide a rationale and be wary that this may reduce the number of eligible studies. Review question: The second objective focuses specifically on viral suppression, whereas the previous one is more broadly treatment outcomes. Do you intend to evaluate effectiveness of interventions for the other outcomes as well, or only VS? You should indicate if so and explain why if only VS. Inclusion criteria: Does not specifically state that you will only look at studies for adolescents who were virally unsuppressed, which is what you've indicated as the novel contribution for this SR. Would be good to clarify how exactly you have designed the SR to focus on virally unsuppressed ALHIV. Will you be including all quantitative studies? If not, what do you define as "comparisons"—does it have to be between study arms (i.e. intervention vs. control) or can it be within the same cohort (i.e. before vs. after intervention)? What is the rationale for only including studies published from 2010? What do you define as grey literature that you're including in the study, if you're excluding conference abstracts and presentations among others? Including children <10 years old and >19 years old does not match the inclusion criteria stated or the scope of the review. The inclusion of >19 year olds has been done in previous SRs that focus on adolescents and young people/youth, but this is one of the critiques you provide in the study rationale. How will you approach inclusion/data extraction if you find studies that test interventions on children and young adults? What if they are interventions for 18 years and older? What if they are broadly available to all PLHIV and include ALHIV? PICOT It's unclear from how this is written if you're also considering DSD to be an intervention. What do you define as an intervention vs. health policy change like decentralizing HIV care? This seems like quite a broad scope. Also see previous note on the study rationale's comment on interventions outside the clinic. How feasible is it to expect that interventions are only targeting virally unsuppressed adolescents? Were any studies of this nature included in previous SRs? It seems like this is narrow beyond the point of being practical (i.e. interventions can be targeted to ALHIV who have a history of challenges with VS or poor retention in care, but that doesn't necessarily mean they are VS). Viral loads are

	also not always recent/available, so this inclusion criteria may actually preclude evaluation of many vulnerable adolescents. See previous note on how comparisons are defined. Outcome measures Retention variables – Transfer out of care doesn't necessarily reflect a lack of retention in care and should be considered separately. Mortality is also an entirely different outcome when available. Should consider doing a sub-analysis on that if possible. Viral suppression-- How will you approach meta-analyses for studies that use different cut-offs for VS? Will you run sensitivity analyses? Search strategy Would also include LTFU among search terms for retention in care. Data analysis: GRADE provides an overall synthesis of the quality of the evidence in the SR. Consider using additional tools to systematically evaluate the quality of each individual included study, such as the CASP checklists and ROBINS-I for assessment of bias. The inclusion criteria stated here is different than previously stated—including adolescents "who may have had adherence challenges." This seems like including adolescents with a history of viral unsuppression but may be currently suppressed. The inclusion criteria should be very clear and consistent.

REVIEWER	Rotheram-Borus, Mary UCLA, Psychiatry
REVIEW RETURNED	16-Mar-2021

GENERAL COMMENTS	BMJ Open Protocol paper: Interventions for improving treatment outcomes in adolescents on ART with unsupressed viral loads: a systematic review protocol This is a high priority area for HIV research. The authors have included a review of the literature that only focuses on low and middle income countries (LMIC) - however, that is the target of their review. They have accurately cited at least three other reviews of this area, the latest being in 2019. They expect to review articles from 2010-currently, published articles. There are several issues that I am concerned about and which need to be more clearly laid out to the reader prior to accepting this protocol paper for publication. Rates of viral suppression have risen markedly over the last five years, as treatment regimens have improved dramatically. Viral suppression is likely to appear at the rate they are setting their criteria (<1000 copies/ml). The authors need to consider the change in treatment regimens as a major determinant of viral suppression. The authors also may want to reconsider the level at which they consider viral suppression. Almost all adolescents are

	now in the range of <100 copies/ml if on recent regimens, even when non-adherent. This is an issue that, if included in their review, would really expand the literature. The previous reviews by others did not address this issue. However, in high income countries, the increase in viral suppression is unlikely to reflect increased adherence, but changes in regimens. Is there a similar process in LMIC? Second, not until you get to the nitty-gritty of their methods do they indicate they will include perinatally infected adolescents and behaviorally infected youth. These are really two different populations with two different histories. Perinatally infected children had parents who would have significantly influenced their habits of daily living, health care uptake, ets. They are likely to respond to very different interventions. Behaviorally infected youth have histories which are likely to shape and influence their interventions and responses to interventions. The review needs to distinguish and highlight the differences in these populations. This proposed review is almost silent to the issue until the Methods section. They chose an age range of 10-19 years. This is justifiable. It will make their results incompatible with the reviews conducted in high income countries. The funding streams in high income countries, not just the U.S., will cut service reimbursement at different ages than 19 years. In South Africa, there will be a break at 18 years. The age breaks may or should be an emerging finding of this review. The classification of the interventions is basically health system, mHealth, behavioral, family/not family. The types of drugs, settings (e.g., school-based -/not), incentive systems (e.g., conditional cash transfers) are broader than what is discussed. Dose also matters and could/should be a salient dimension I this review. The high rates of infection among adolescents warrants this review. I encourage the authors to consider some of these issues.

VERSION 1 – AUTHOR RESPONSE

Reviewer comments	**Revisions made**
Reviewer 1	
Abstract: Some of the language should be made more precise as you write up the review, e.g. Line 15: "unsuppressed adolescents" should be made more specifically "virally unsuppressed adolescents"	Edited as recommended to "virally unsuppressed adolescents"

Methods: For a more comprehensive review, I would also look into databases of grey literature. To my knowledge, the six databases listed do not include grey literature. What tools will you use to systematically assess bias in included studies? What about tools for assessing quality of included studies? Will this SR be focusing on ALHIV care globally, or will it focus on ALHIV in a specific region? Should clarify upfront.	We have added grey literature and citation searches. Added in abstract that focus is on ALHIV globally Added CASP checklists and ROBINS-I tool for quality and bias assessments
Strengths/limitations: When you say "specifically for adolescents with unsuppressed viral loads," does this mean your inclusion criteria will require all study participants to be unsuppressed?	We have broadened the review to not only be limited to unsuppressed adolescents.
Intro The debate around threshold-setting for VS has also been tied to the sensitivity of available tests for detecting viral loads, which may differ across and within countries.	Added sentence on VS assay limit of detection
It seems that the Casale et al and Lindsey et al papers would address the limitations of the MacPherson et al. paper. It would be good to specifically indicate where your SR would add new evidence between these two, among the other SRs listed. When looking at the existing SRs, how many of the participants in the included studies were unsuppressed at baseline? The introduction is setting me up to believe that you will only look at studies where all participants were unsuppressed at baseline, which may be *too* narrow of a focus and likely to yield low number of studies.	We have broadened the SR to not only focus on unsuppressed adolescents, however the new evidence is around recent papers with particular focus on unsuppressed PLHIV, including adolescents e.g. Diress *et al*, 2020, Nasuuna *et al*, 2018, Bvochora *et al*, 2019, Mavhu *et al*, 2020, which have implemented different models of Enhanced Adherence Counseling (EAC) and Community Adolescents Treatment Supporters (CATS) interventions.
Rationale for review: It seems that the reason for the low number of studies included in previous SRs is simply that there is a lack of sufficient studies on this topic. This is also the reason why many SRs have expanded eligibility criteria to include youth up to the age of 24. What does your SR do beyond these existing ones? Has there been a sufficient amount of time since the most recent SR that there is new data worth synthesizing? It would	We have added the above-mentioned recent studies in the intro and referenced recommendations from previous SRs to review studies focusing on the 10-19 age group, if the specific gaps for this age group are to be addressed. Age disaggregation will be considered for studies including older youths

also seem important to include interventions across all levels—clinic, community, and household. But from this paragraph, it sounds like you're making the case to reduce that scope. If so, you should provide a rationale and be wary that this may reduce the number of eligible studies.	Additions are also to look at interventions across all levels i.e. clinic, community and household
Review question: The second objective focuses specifically on viral suppression, whereas the previous one is more broadly treatment outcomes. Do you intend to evaluate effectiveness of interventions for the other outcomes as well, or only VS? You should indicate if so and explain why if only VS.	The review questions have been edited to reflect the recommended broader scope of the SR
Inclusion criteria: Does not specifically state that you will only look at studies for adolescents who were virally unsuppressed, which is what you've indicated as the novel contribution for this SR. Would be good to clarify how exactly you have designed the SR to focus on virally unsuppressed ALHIV.	We have broadened the review to not only be limited to unsuppressed adolescents.
Will you be including all quantitative studies? If not, what do you define as "comparisons"—does it have to be between study arms (i.e. intervention vs. control) or can it be within the same cohort (i.e. before vs. after intervention)?	The comparisons will include pre-/post- or multi-arm comparison groups.
What is the rationale for only including studies published from 2010?	We considered the number of SRs previously done and thought the last 10 years would provide a reasonable quantity (and quality) of recent interventions to include at this time.
What do you define as grey literature that you're including in the study, if you're excluding conference abstracts and presentations among others?	
Including children <10 years old and >19 years old does not match the inclusion criteria stated or the scope of the review. The inclusion of >19-year olds has been done in previous SRs that focus on adolescents and young people/youth, but this is one of the critiques you provide in the study rationale. How will you approach inclusion/data extraction if you find studies that test interventions on children and young adults? What if they are interventions for 18 years and older? What if they are broadly available to all PLHIV and include ALHIV?	Age disaggregation will be performed where there are age overlaps and the authors have provided age-disaggregated data.

PICOT It's unclear from how this is written if you're also considering DSD to be an intervention. What do you define as an intervention vs. health policy change like decentralizing HIV care? This seems like quite a broad scope. Also see previous note on the study rationale's comment on interventions outside the clinic.	Decentralization of HIV care is considered a mode of differentiated HIV care, so reports on evaluations of DSD models will be considered. Interventions outside of the clinic are included e.g. community-based and home-based interventions. We have added to the PICOT table
How feasible is it to expect that interventions are only targeting virally unsuppressed adolescents? Were any studies of this nature included in previous SRs? It seems like this is narrow beyond the point of being practical (i.e. interventions can be targeted to ALHIV who have a history of challenges with VS or poor retention in care, but that doesn't necessarily mean they are VS). Viral loads are also not always recent/available, so this inclusion criteria may actually preclude evaluation of many vulnerable adolescents.	We have broadened the review to not only be limited to unsuppressed adolescents.
Outcome measures Retention variables – Transfer out of care doesn't necessarily reflect a lack of retention in care and should be considered separately. Mortality is also an entirely different outcome when available. Should consider doing a sub-analysis on that if possible.	A sub-analysis of the different categories of adolescents who are not still engaged in care will be performed.
Viral suppression-- How will you approach meta-analyses for studies that use different cut-offs for VS? Will you run sensitivity analyses?	Where different cut-offs for viral load suppression are used, we will run sensitivity analyses
Search strategy Would also include LTFU among search terms for retention in care.	We have included retention in the search terms
Data analysis: GRADE provides an overall synthesis of the quality of the evidence in the SR. Consider using additional tools to systematically evaluate the quality of each individual included study, such as the CASP checklists and ROBINS-I for assessment of bias.	Added CASP checklists and ROBINS-I tool for quality and bias assessments
The inclusion criteria stated here is different than previously stated—including adolescents "who may have had adherence challenges." This seems like including adolescents with a history of viral unsuppression but may be currently suppressed. The inclusion criteria should be very clear and consistent.	We have aligned the inclusion criteria and the data analysis section to include other adolescents, not limited to unsuppressed ALHIV only.

Reviewer 2	
Rates of viral suppression have risen markedly over the last five years, as treatment regimens have improved dramatically. Viral suppression is likely to appear at the rate they are setting their criteria (<1000 copies/ml). The authors need to consider the change in treatment regimens as a major determinant of viral suppression. The authors also may want to reconsider the level at which they consider viral suppression. Almost all adolescents are now in the range of <100 copies/ml if on recent regimens, even when non-adherent. This is an issue that, if included in their review, would really expand the literature. The previous reviews by others did not address this issue. However, in high income countries, the increase in viral suppression is unlikely to reflect increased adherence, but changes in regimens. Is there a similar process in LMIC?	Yes, there are efforts to transition all eligible clients on ART to the more recent regimens, in Namibia the transition to DTG-based regimens was rolled out in 2019. We are not sure if there has been any published studies on this by 2020 (SR timeline), but we will include any articles considering effects of more robust ART regimens found through the review. We will also consider different cut-off points for viral suppression from the different studies as some studies may use lower cut-off points than the WHO 1000 copies/ml.
Second, not until you get to the nitty-gritty of their methods do they indicate they will include perinatally infected adolescents and behaviorally infected youth. These are really two different populations with two different histories. Perinatally infected children had parents who would have significantly influenced their habits of daily living, health care uptake, ets. They are likely to respond to very different interventions. Behaviorally infected youth have histories which are likely to shape and influence their interventions and responses to interventions. The review needs to distinguish and highlight the differences in these populations. This proposed review is almost silent to the issue until the Methods section.	We have added a reference to the differences between the perinatally and behaviourally infected ALHIV upfront in the introduction.
They chose an age range of 10-19 years. This is justifiable. It will make their results incompatible with the reviews conducted in high income countries. The funding streams in high income countries, not just the U.S., will cut service reimbursement at different ages than 19 years. In South Africa, there will be a break at 18 years. The age breaks may or should be an emerging finding of this review.	This will be part of this review, we will provide age-disaggregated analyses where possible.

The classification of the interventions is basically health system, mHealth, behavioral, family/not family. The types of drugs, settings (e.g., school-based -/not), incentive systems (e.g., conditional cash transfers) are broader than what is discussed. Dose also matters and could/should be a salient dimension I this review.	We will consider all interventions reported in studies identified through the SR, including types of medication and dose.

VERSION 2 – REVIEW

REVIEWER	Haghighat, Roxanna University of Oxford, Social Policy and Intervention
REVIEW RETURNED	12-Jun-2021

GENERAL COMMENTS	As noted by the other reviewer, it would be appropriate to analyse results from the SR by mode of infection among the adolescents included in meta-analysis.

REVIEWER	Foster, Caroline Imperial College London
REVIEW RETURNED	10-Jun-2021

GENERAL COMMENTS	The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

VERSION 2 – AUTHOR RESPONSE

Reviewer comments	**Revisions made**
Reviewer 1	
Introduction Inserted comments on Peer Review document	All suggested edits added to introduction, research question, PICO table, outcome measures
Rephrase the WHO definition of adolescents	Words “children and young adults removed
Consider clarifying if related studies reported adherence only or to actual viral load data, and	Outcomes reported clarified in text and a PrEP study referenced on limitations of self-report among younger age groups

the limitations of self-report particularly in adolescents	
Review question: The review question does not include failure of viral suppression which is outlined as the unique focus as compared to previous reviews	The review questions have been edited to reflect the focus on failure of viral suppression as the focus
PICOT. Population should reflect focus on VS, to align with the review question	Added focus on VS among the 10-19 age group
Outcome measures Viral load as the primary outcome should be first on the outcomes list/definitions	Moved viral load measures to the top of the list
It might be useful for the reader to acknowledge the limitations of adherence tools in correlation with VS. Many adolescents want to be seen to be doing the right thing, and therefore over self-report adherence	We have added limitations of adherence tools vs VS, and socially desirable responses from adolescents.
How will the retention sub-analysis be performed given by definition the lack of data on this group	A sub-analysis of the different categories of adolescents who are not still engaged in care will be performed if possible (when different retention variables are reported).
Search strategy Consider adding “youth”, since it’s too late would add as a potential limitation	Noted, we will add this as a limitation to the search
Data analysis: As noted by the other reviewer, it would be appropriate to analyse results from the SR by mode of infection among the adolescents included in meta-analysis.	We have added the disaggregated analysis to the Data analysis section
Reviewer 2	
It is clearer from the authors responses to reviewers than in the article that the focus will be VS but broader outcomes reported - the latter similar to previous published reviews but the introduction still heavily focuses on VS. As such the introduction needs to be modified to make clear earlier that whilst VS is the main focus broader outcomes will be considered including adherence, retention, mortality, loss to follow up etc.	We have added a statement in the introduction to emphasize that although the primary focus is VS, other outcomes to be considered are adherence and retention in care.

It would be extremely helpful to the reader very early in the introduction to have a clear outline of the population (age 10-19 years, perinatally and behaviourally acquired HIV infection, high middle and low income settings), and then interventions and outcomes assessed ; VS, retention, adherence etc.	We have added the population to the first paragraph of the introduction
Similarly if the focus is on VS but in the context of wider outcomes then this should be clear in the review question which currently does not mention the existing primary outcome (VS).	The review questions have been edited to reflect the focus on failure of viral suppression as the focus
Inclusion criteria - clarify that both perinatal and behaviourally infected adolescents included and consider adding to 7. All countries, high middle and low income settings 2010-2020	Recommended edits and additions made to the inclusion criteria